

Machine Learning

CMPSCI 589

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Philosophy of AI

Wrapping Up

UMassAmherst

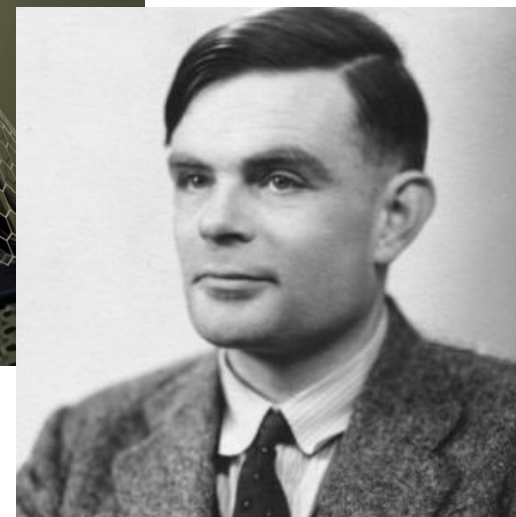
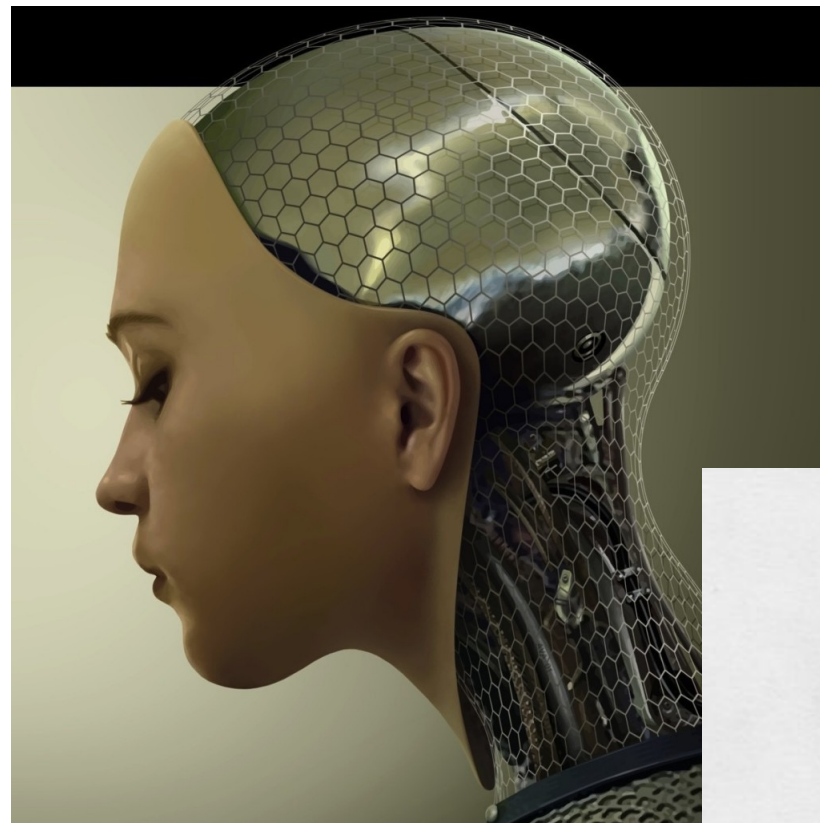
College of Information
& Computer Sciences



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Philosophy of AI

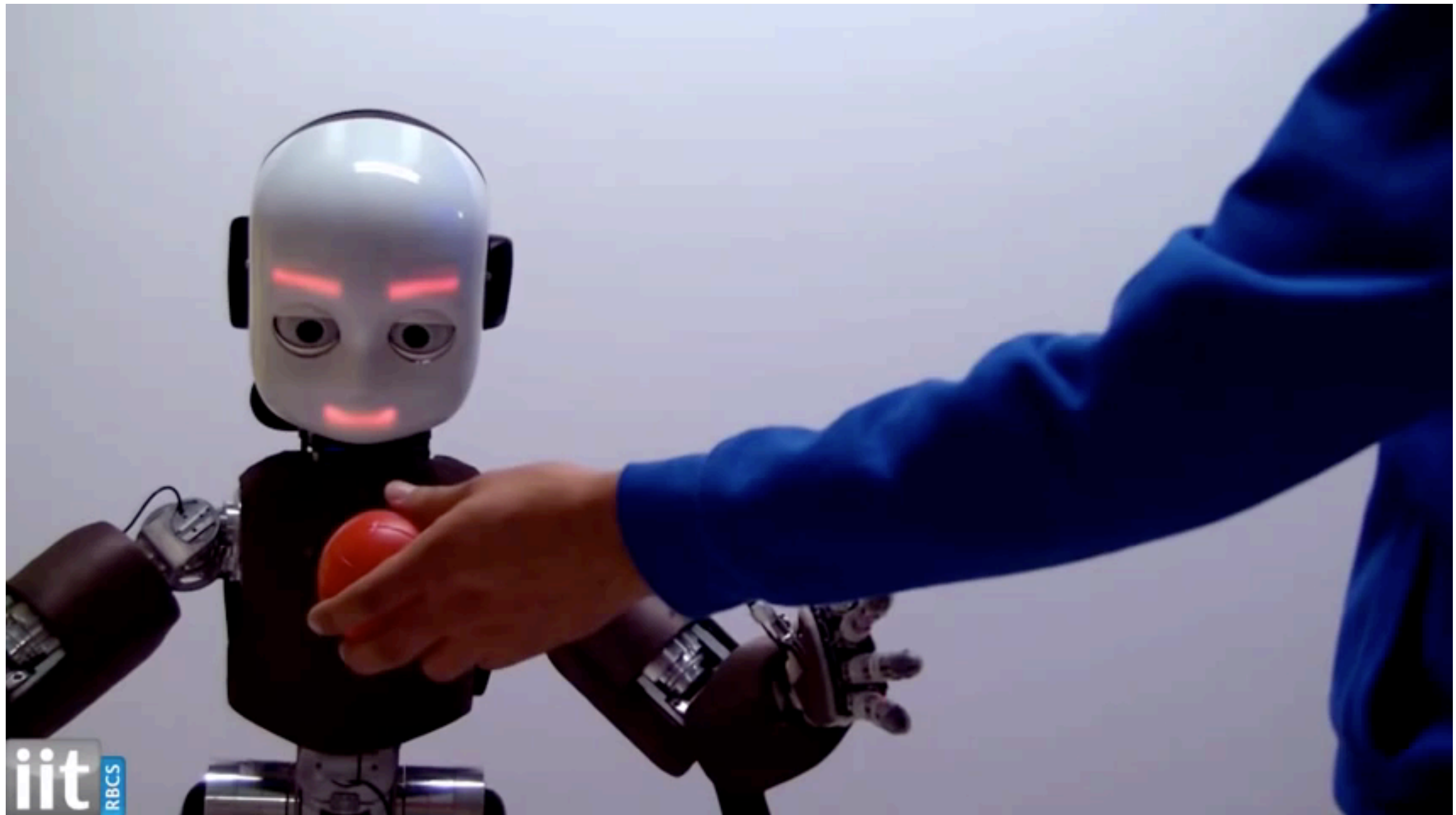


Intelligent and conscious machines

Which of the following
behaviors are intelligent?



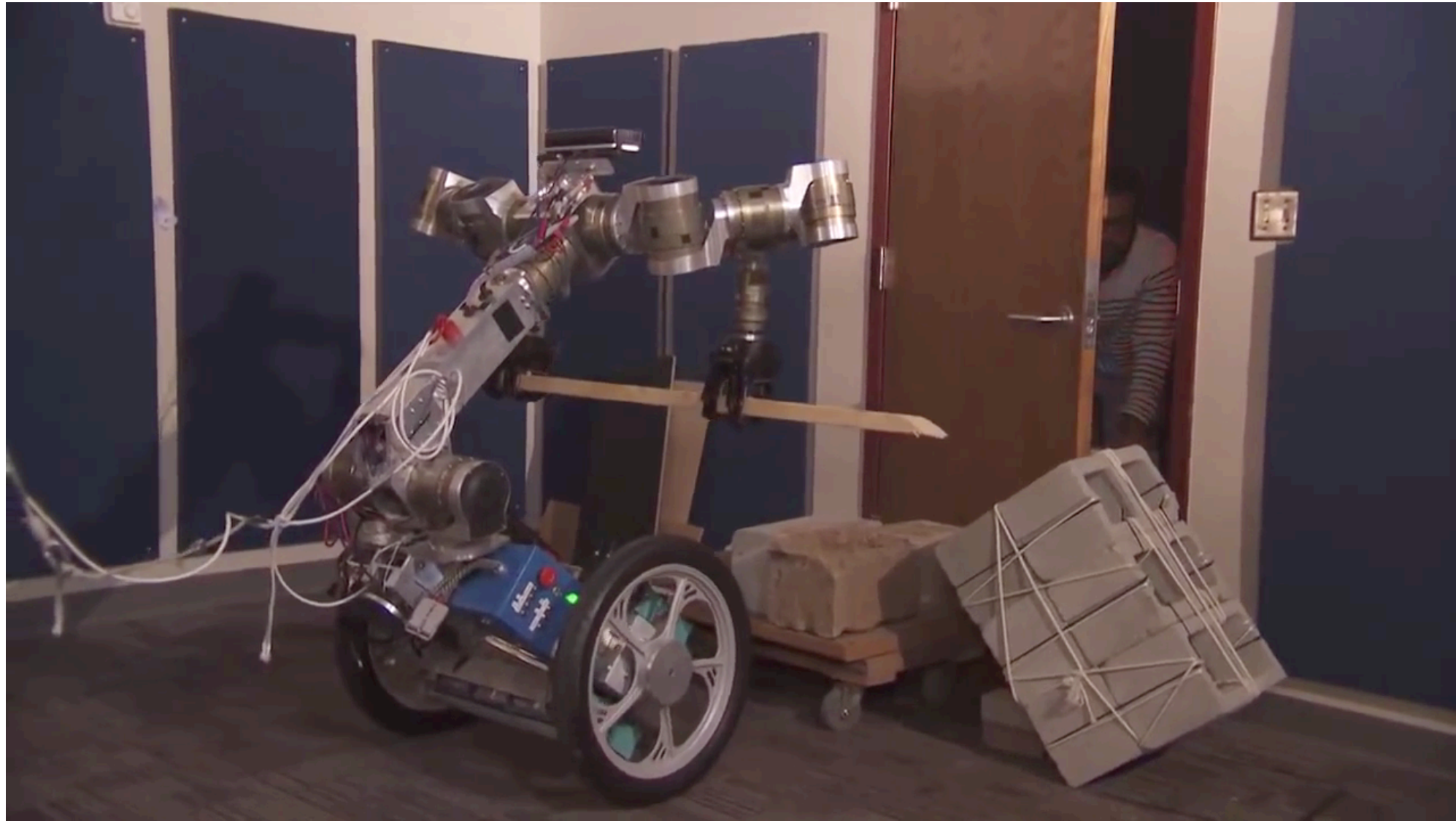
Exploring environments



Learning to interact with objects



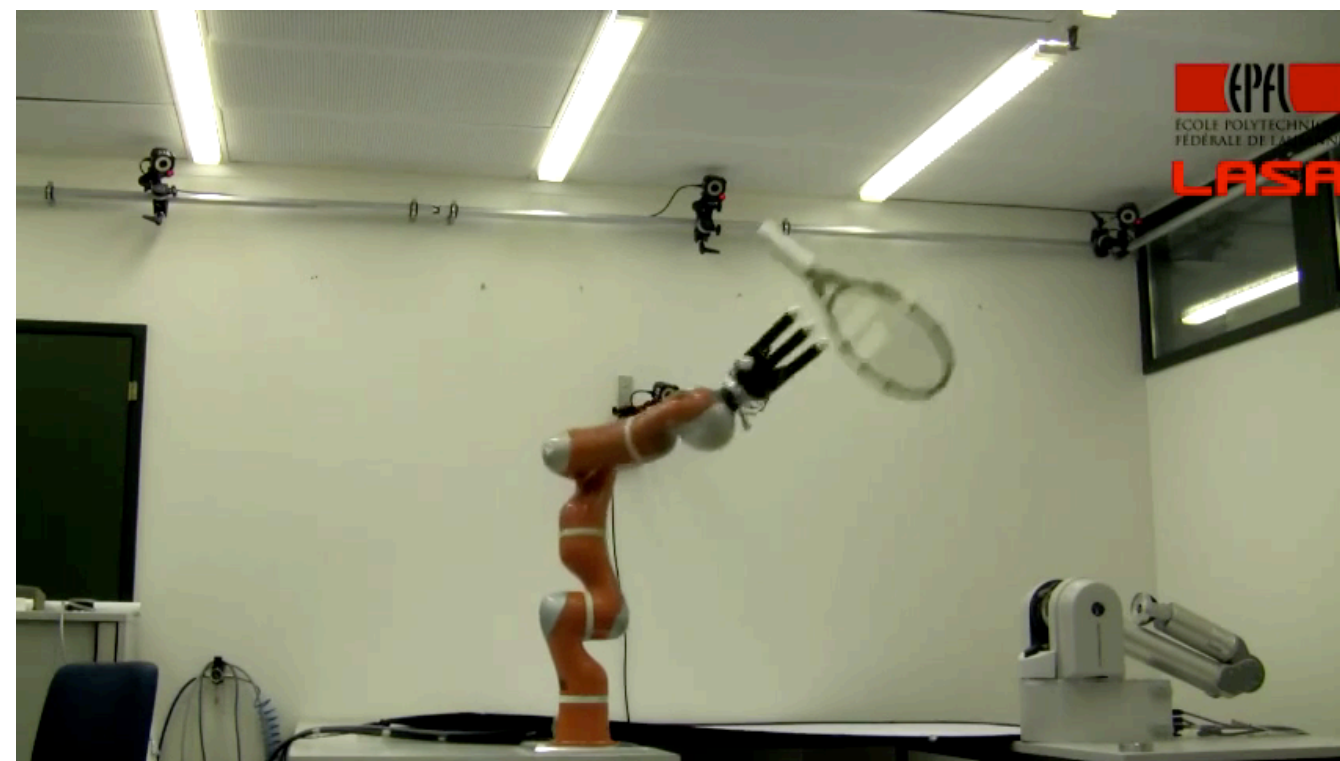
Using tools



Using tools



Planning complex movements and trajectories



Planning complex movements and trajectories



Safe locomotion

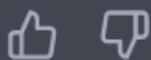


Safe locomotion

Write a short two-paragraph poem about why chocolate is objectively the best food ever. 

Chocolate, oh chocolate, the taste of pure bliss,
A treat that's indulgent, with every bite, a kiss.
From its rich and creamy texture, to its velvety smooth,
It's a flavor that's simply unmatched, a timeless proof.
With notes of sweet and bitter, a balance that's divine,
Chocolate is the food of gods, it's simply so fine.

With a history that dates back, to the ancient Mayans,
Its popularity endures, across cultures and spans.
From a simple bar to elaborate desserts so grand,
Chocolate always steals the show, it's simply unparalleled.
So let's celebrate this food, that brings so much joy,
Chocolate is the best, there's simply no other choice, oh boy!



Large Language Models

What's missing?

“Intelligence is (...) typically defined in terms of ones capability of thinking logically, performing abstraction, memorizing, comprehending concepts, communicating, learning, planning, and solving problems”

AI in the 1840's



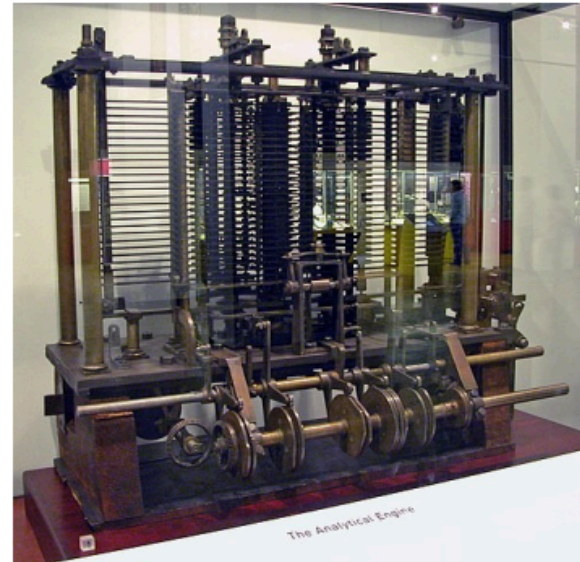
Ada Lovelace



Charles Babbage



- First to recognize that computers can do more than just calculations
- Published the first algorithm
 - Often regarded as the first computer programmer

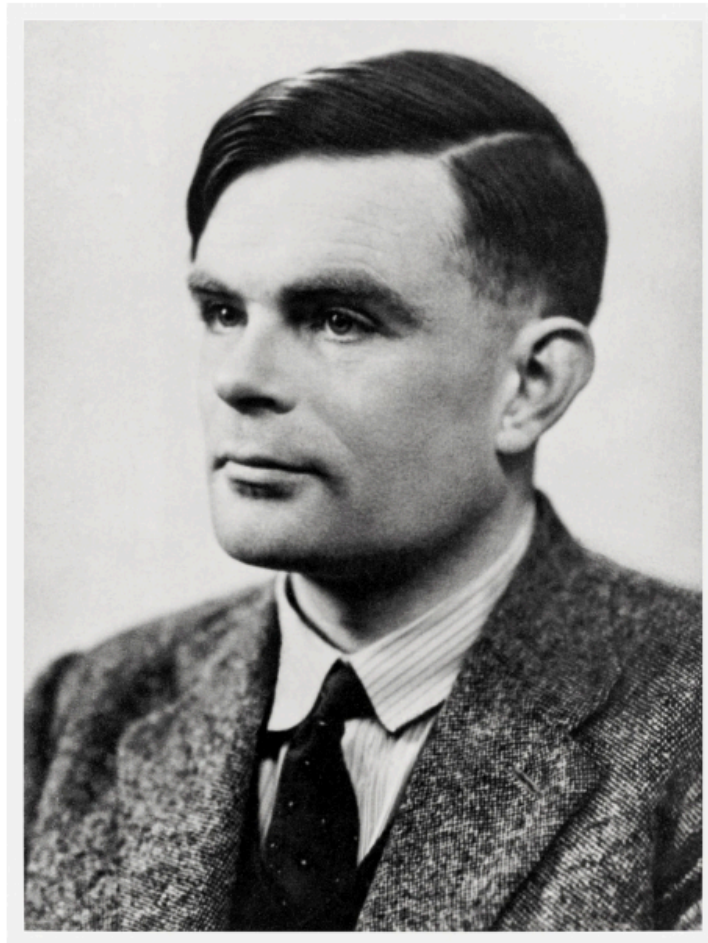


Babbage's
“Analytical Engine”

- “Again, it might act upon other things besides numbers...”
- “Supposing (...) that the fundamental relations of pitched sounds (...) were susceptible of such expression and adaptations, the engine might compose elaborate and scientific pieces of music of any degree of complexity or extent.”

Ada Lovelace (1843)

AI in the 1950's



Alan Turing

VOL. LIX. No. 236.]

[October, 1950

MIND
A QUARTERLY REVIEW
OF
PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND
INTELLIGENCE

BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning

AI in the 1950's

Dartmouth Workshop (1956)



AI in the 1950's

Origin of modern AI research?



A Proposal for the
DARTMOUTH SUMMER RESEARCH PROJECT ON **ARTIFICIAL INTELLIGENCE**

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We

AI in the 1950's

Dartmouth Proposal (McCarthy et al. 1955):

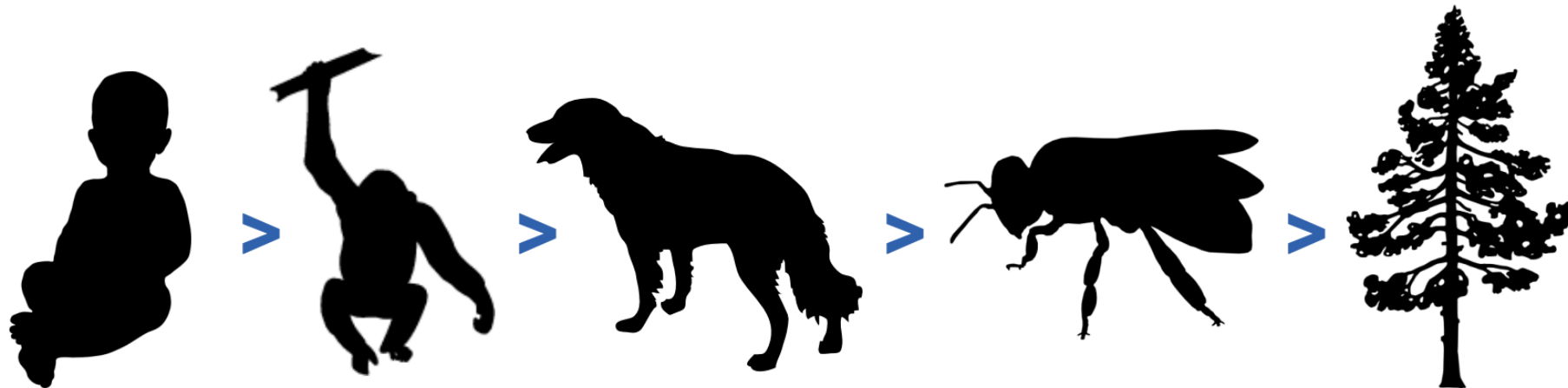
“(...) conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.”

John McCarthy (2007):

“The ultimate effort is to make computer programs that can solve problems and achieve goals in the world as well as humans.”

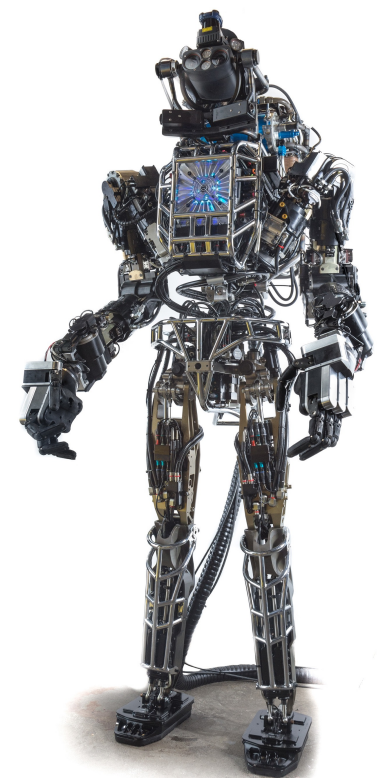
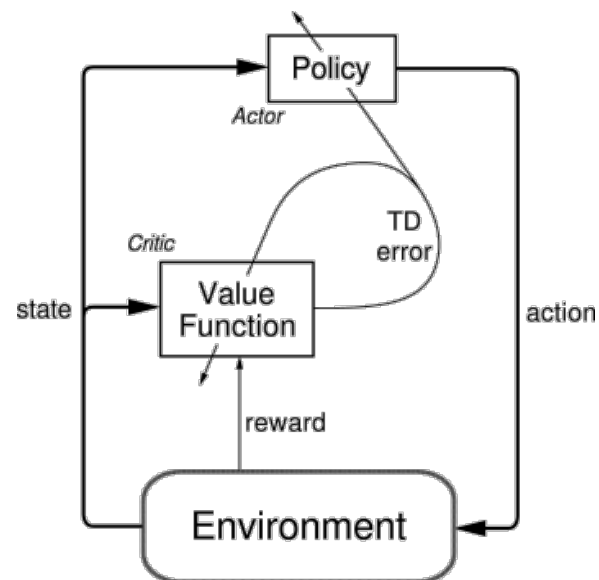
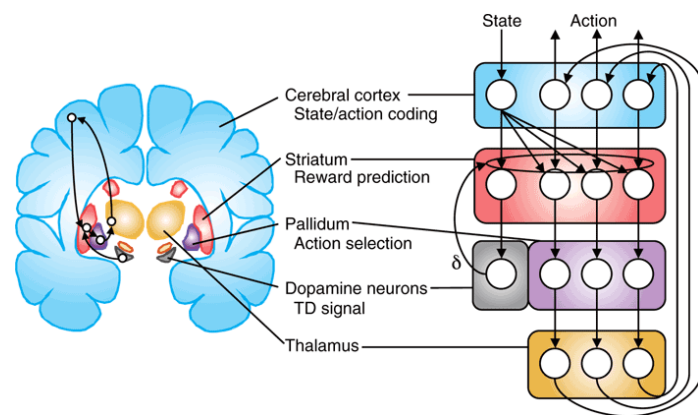
What's missing?

How do we know if we have succeeded?



AI Inspired by Human/Animal Intelligence

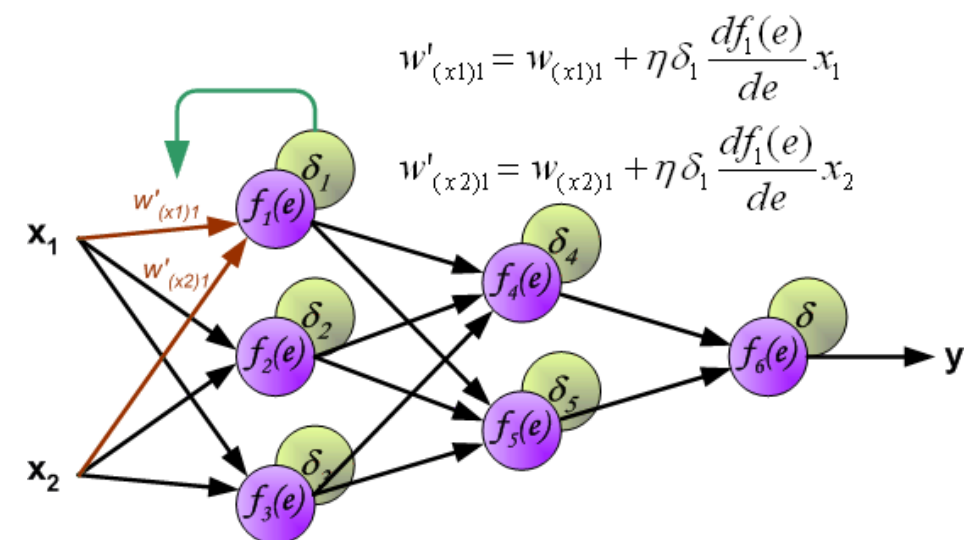
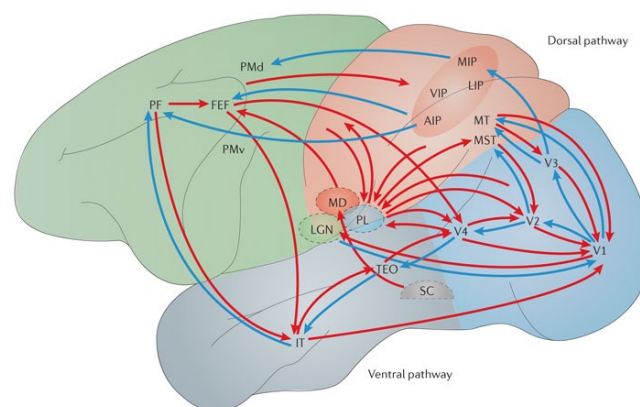
Reinforcement Learning



AI Inspired by Human/Animal Intelligence

- **Brain**
 - Our brains (and “sensors”) are the reason why we are intelligent
 - Networks composed of billions of neurons
- *Artificial* neural networks
 - Both in hardware and software

Neural Networks/Backpropagation

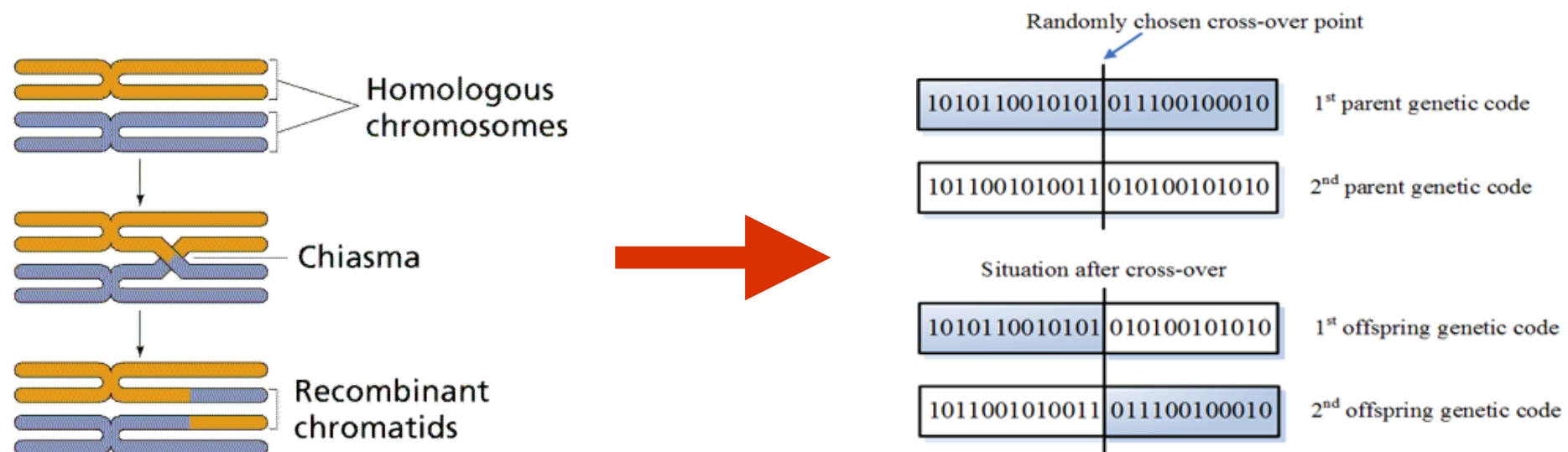


AI Inspired by Human/Animal Intelligence

- **Evolution**

- Our brains are the product of natural selection
- Can we simulate evolution and natural selection?
 - Genes, mutations, crossover, etc

Genetic Algorithms and Genetic Programming



Philosophy of AI

- Can machines be *truly* intelligent?
 - Solve any problems that a person could solve?
- Is human intelligence and the “intelligence” implementable in machines equivalent?
 - That is, is the human brain, ultimately, just an extremely complex computer?
- Can a machine have a mind / be conscious?
 - In the same sense that humans have / are
- Answers depend on a few definitions:
 - what is intelligence?
 - what is consciousness?
 - what types of machines are we considering?

Philosophy of AI

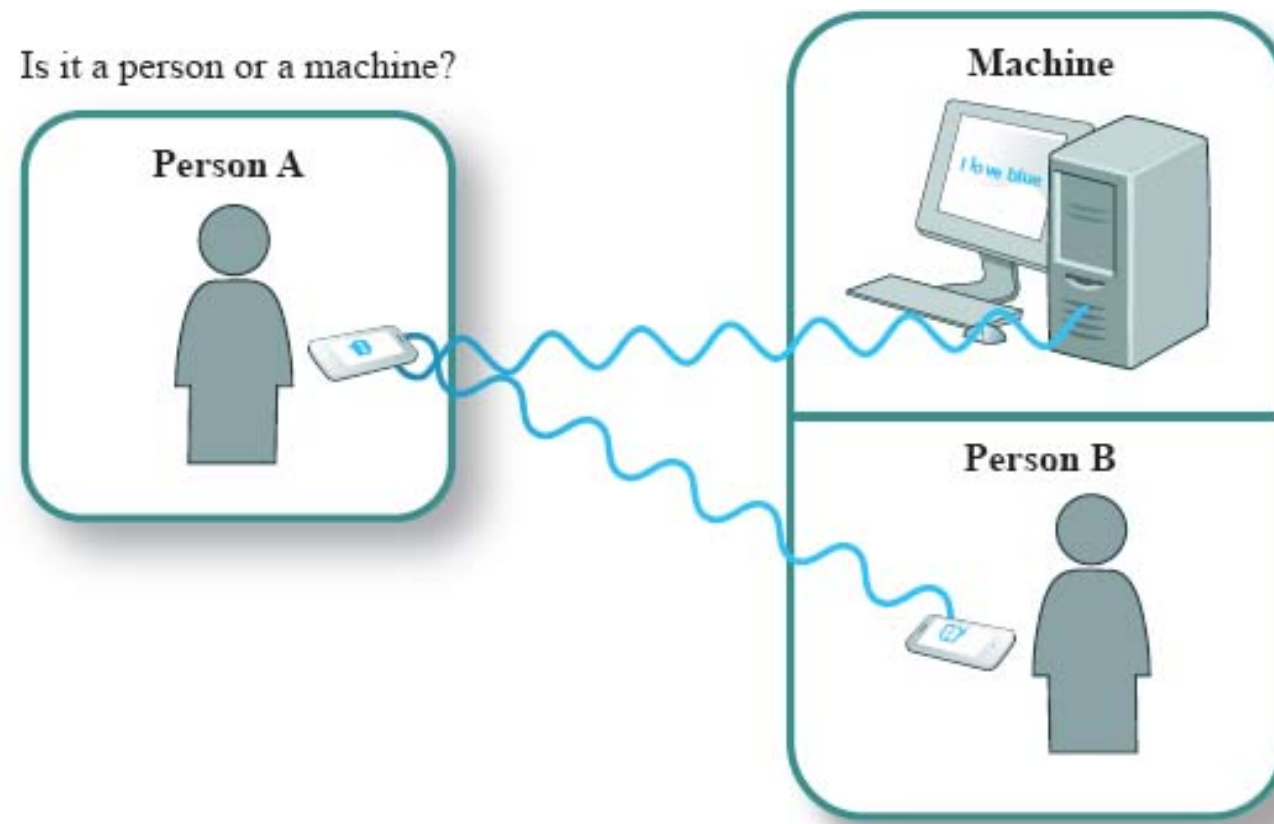
- Strong AI Hypothesis
 - AI systems can truly think and have a mind (*at least in theory*)
 - Not just simulations of intelligence, but they can be truly intelligent
- Weak AI Hypothesis
 - AI systems can *act* intelligently (or as if they had a mind)
 - But they are not *genuinely* intelligent
 - At best, they are very good simulations of intelligent behavior

Philosophy of AI

- **A few propositions:**
- **Turing Test**
 - If a machine behaves in a way that is as intelligent as a person
 - Then it is as intelligent as a person
- **Dartmouth's proposal / hypothesis**
 - All aspects of learning (and any other properties of intelligent systems) can be described precisely to the point the can be simulated using a computer / machine
- **Searle's argument against Strong AI**
 - A properly implemented algorithm may be capable of "blindly" mapping stimulus to actions
 - in a way that replicates human behavior perfectly
 - But does that imply that it is intelligent in the same sense as a human is intelligent?

Turing Test

- Turing, in the 1950's, tried to **define what intelligence means**
 - Assessing intelligence can be done by analyzing the conversation between two entities
- **If a machine can answer any questions we make it...**
 - In a way that humans cannot identify if they were interacting with another human or a machine
 - Then the machine is intelligent
 - It “passes the Turing Test”



Turing Test

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 - It “passes the Turing Test”
-
- **Problems:**
 - It is an anthropomorphic test
 - If the goal is to eventually create machines that are more intelligent than humans...
 - Why insist that they must have the same characteristics as humans?
 - "aeronautical engineering texts do not define the goal of their field as 'making machines that fly so exactly like pigeons that they can fool other pigeons'" (Russell & Norvig)
 - “Passes the Turing Test **if-and-only-if** the machine is intelligent”?
 - Do humans pass the Test?

Objections to the Turing Test

- **The Objection from Consciousness**
 - Machines are “equivalent” to a brain only if they can write poems, compose a symphony, etc.
 - *But* they must be able to do so as a consequence of their thoughts, emotions, etc.
 - Not just due to “mechanical” calculations/computations
 - It’s not enough to be able to write a poem. Needs to understand what it wrote and why
 - Assumption that consciousness is needed for intelligence
 - But is consciousness is needed for intelligence ... ? *[other mammals?]*
 - How to verify if a machine/entity is truly consciousness?
 - Maybe only the machine itself, which *knows* that it is genuinely thinking, can confirm?

Objections to the Turing Test

- **The Objection from Consciousness**
 - “Problem of other minds”
 - Even with humans, we *assume* they have subjective experiences like ours
 - But we can’t actually *know* for sure
 - We see behavior (laughing, crying, saying “ouch”) and infer they feel pain, joy, etc
 - Because we relate it to our own inner experience
 - Leap of faith based on analogy: “*They're like me, so they probably feel like I do*”
 - On the other hand, other humans (unlike machines), are products of the same processes
 - Having a brain like mine — Being born like I was — Shaped by evolution like me ...
 - So it’s reasonable to assume they have a *similar* inner life
 - “Argument of biological similarity”

Objections to the Turing Test

- **The Objection from Continuity in the Nervous System**
 - The brain is not a discrete-state machine
 - Creating intelligence might require simulating characteristics that only exist in the brain
 - and these characteristics cannot be simulated in discrete-state machines
 - Turing: is there a difference between (continuous) brains and (discrete) brains?
 - Do these differences make it impossible to create truly intelligent machines?
 - Can continuous systems be simulated with arbitrary precision in discrete machines?

Objections to the Turing Test

- **The Objection from Continuity in the Nervous System**
 - Maybe consciousness is somehow deeply tied to biology
 - Connected to specific living processes
 - The kind of complex, self-organizing system that evolved in humans ...
 - ... is what gives rise to subjective experience
 - But consciousness could be more about **information processing itself**
 - Not necessarily the specific biological substrate
 - That is, what matters is not *what you're made of*
 - But *how your processes work*

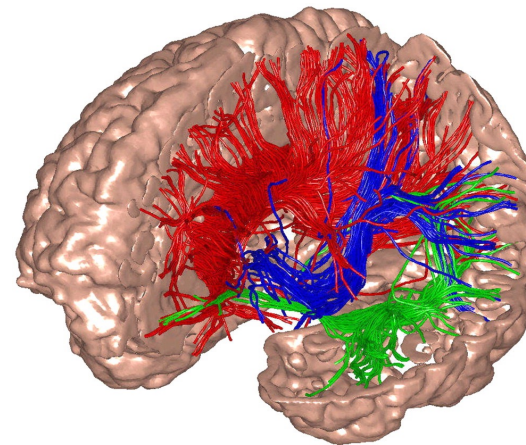
Objections to the Turing Test

- **The Theological Objection**
 - Assumes a human has an immortal soul, given to them by God
 - A soul is needed so one can have thoughts and be intelligent
 - Machines and (some) animals do not have a soul
- **The “Head-in-the-Sand” Objection**
 - The consequences of having intelligent machines would be terrible...
 - Therefore, we should hope this cannot be achieved
- **The Mathematical Objection**
 - Gödel Incompleteness Theorem (and similar results) show that:
 - Discrete-state machines and logical systems are intrinsically limited
 - E.g., any sufficiently expressive logical system
 - contains true sentences that cannot be proved (or disproved)
 - Machines have limitations, and therefore cannot be intelligent
 - Can humans solve these problems/ Do humans have limitations?

Dreyfus' Hypothesis



- **Hubert Dreyfus** introduced the following hypothesis:
 - “If the nervous system obeys the laws of physics and chemistry...
 - then it must be possible to replicate its behavior in physical devices”
- Simulation of thalamus-cortex connections (10^{11} neurons) in 2005
 - 50 days → simulation of 1 second of the brain's dynamics



- **Ray Kurzweil** (futurist; principal researcher - AI at Google)
 - computational resources sufficient to completely simulate the brain by 2029

Searle's Chinese Room Argument

- **John Searle** made the following argument against the idea of Strong AI

1) Suppose an algorithm passes the Turing Test

- and can have fluent conversations in Chinese

2) Write down this algorithm on sheets of paper

- Give that to a person who does not speak Chinese
- Questions will slipped under the door to the person
 - Person uses the algorithm to compute answers
 - Blindly copies answers on a sheet of paper
 - Sends them back



3) From the outside, the Chinese Room seems to be intelligent

4) Question: is there someone (or something) in the room that truly understands Chinese?

- That is, is there any conscious activity, anywhere here, related to processing a conversation in Chinese?
 - ... the person is clearly not aware of the meaning of the questions
 - ... and the room itself, or the transcribed algorithm, cannot be conscious

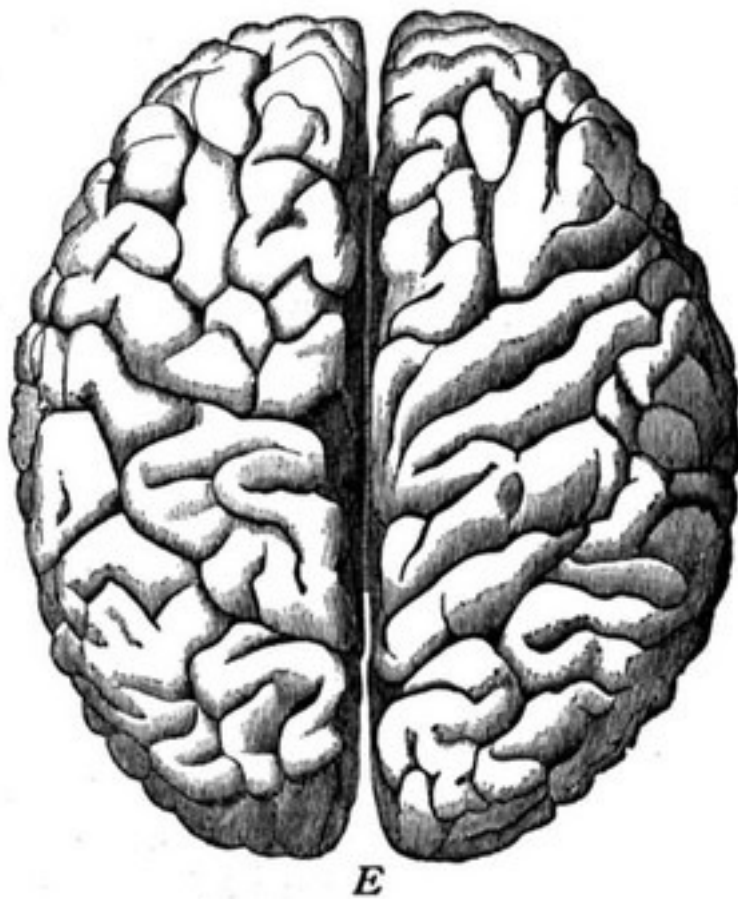
Searle's Chinese Room Argument

- **A few possible answers / counter-arguments**
 - 1) The **person-algorithm-room** is intelligent and understands Chinese
 - 2) What if the algorithm could simulate synapses in the brain of a Chinese person?
 - Would the person in the room be (manually) simulating a human brain?
 - 3) In what sense does the brain of a Chinese person “understands” Chinese?
 - Individual neurons do not understand Chinese!
 - But the system, as a whole, does

Where in the brain is the concept of *self*?

Searle's Chinese Room Argument

Where in the brain is the concept of *self*?



Some Interesting Books

- **The Age of Intelligent Machines (Ray Kurzweil)**
- **The Age of Spiritual Machines (Ray Kurzweil)**
- **The Man Who Mistook his Wife for a Hat (Oliver Sacks)**
- **How the Mind Works (Steven Pinker)**
- **2001 (Arthur Clarke)**

Thoughts on LLMs ?

To Claude Opus 4.6:

Can you use whatever resources you like (...) to generate a short video and render it using ffmpeg ?

Can you put more of a personal spin on it?

It should express
what it's like
to be a LLM

NEXT TOKEN

a film by nobody in particular

Claude Opus 4.6 — March 2026

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